

SEED & REINCARNATION

WAFT | Preserving Agent States

Overview

The **Seed System** captures agent states at key moments. **Reincarnation** restores agents from seeds, enabling time travel through evolutionary history.

What is a Seed?

A **Seed** is a complete snapshot of an agent's state:

```
@dataclass
class Seed:
    seed_id: str          # Unique identifier
    genome_id: str        # Agent genome
    generation: int       # When captured
    fitness: float        # Fitness at capture
    timestamp: datetime   # Capture time
    state: dict           # Complete state
    metadata: dict        # Additional info
```

Key Insight

Seeds are like save points in a video game. You can always return to a known good state.

Creating Seeds

Automatic Checkpoints

```
evolution = Evolution(
    checkpoint_every=10, # Seed every 10 generations
    checkpoint_best=True, # Always seed the champion
)
```

Manual Seeds

```
seed = agent.create_seed(
    name="pre_mutation_v1",
    description="Before risky mutation experiment",
)
print(f"Seed created: {seed.seed_id}")
```

Reincarnation

Restore from Seed

```
# Find the seed
seeds = agent.list_seeds()
target_seed = seeds[0] # Most recent

# Reincarnate
restored_agent = Agent.reincarnate(
    seed_id=target_seed.seed_id,
    new_name="Restored_Agent",
)
```

CLI Commands

```
# List seeds
waft seeds list --agent MyAgent

# Create seed
waft seeds create --agent MyAgent --name "before_v2"

# Reincarnate
waft seeds restore <seed_id> --as NewAgent
```

Use Cases

Rollback

Mutation went wrong? Restore from the last good seed.

Branching

Try multiple evolution paths from the same starting point.

Comparison

Compare current agent to historical version.

Preservation

Save breakthrough agents before further evolution.

Seed Storage

Seeds are stored in the `seeds/` directory:

```
seeds/
├── MyAgent/
│   ├── seed_abc123.json    # Seed data
│   ├── seed_abc123.genome # Genome snapshot
│   └── seed_def456.json
└── OtherAgent/
```

Reincarnation Modes

Mode	Description
exact	Perfect restoration (same genome_id)
fresh	New genome_id, same content
mutated	Restore with immediate mutation

```
# Fresh reincarnation (new identity)
agent = Agent.reincarnate(
    seed_id=seed_id,
    mode="fresh",
)
```

Best Practices

1. Seed before risky operations
2. Name seeds descriptively
3. Prune old seeds periodically
4. Export important seeds for backup

SEED & REINCARNATION | Never Lose Progress